

2. Considering the boundary condition described in Eq. (5.16):

$$\Phi(\phi) = \Phi(\phi + 2\pi), \quad (\text{A.133})$$

we find that $n = \pm 1, \pm 2, \dots$, where $n = 0$ is not considered because it leads to a trivial solution. Using Eq. (A.131), where $[E - V(r)] = p^2/2m$, we find:

$$n^2 = \frac{2mr^2}{\hbar^2} \frac{p^2}{2m} \Rightarrow L = n\hbar. \quad (\text{A.134})$$

3. Considering the Virial Theorem (Theorem 2.2) and the radial Coulomb potential, we have:

$$2\langle\hat{T}\rangle = -\langle V(\hat{r})\rangle. \quad (\text{A.135})$$

However, $E = \langle\hat{T}\rangle + \langle V(\hat{r})\rangle$ and therefore $E = 1/2\langle V(\hat{r})\rangle$. The radius is constant and therefore it is possible to write: $\langle V(\hat{r})\rangle = V(\hat{r})$. All this information into Eq. (A.131) yields:

$$r = a_0 n^2, \quad \text{where} \quad a_0 = \frac{4\pi\epsilon_0\hbar^2}{e^2m} \quad (\text{A.136})$$

is the Bohr radius.

4. Considering $E = 1/2\langle V(\hat{r})\rangle$ we find:

$$E = -\frac{E_0}{n^2}, \quad \text{where} \quad E_0 = \frac{\hbar^2}{2ma_0^2} \quad (\text{A.137})$$

is the ionization energy of the studied atom.

5. This problem reproduces the results of the Bohr atom.

A.6 Chapter 6

Conceptual exercises

Solution 6.1. This commutation relation highlights that it is impossible to precisely determine two components of angular momentum at the same time. Measuring one component causes the other two to become uncertain – a consequence of the uncertainty principle. Mathematically, when these components are represented as matrices, they cannot be diagonalized at once. A basis that diagonalizes one component ensures the other two remain non-diagonal.

Solution 6.2. The variance represents the dispersion of a given distribution. Considering the commutation relation of angular momentum, it is not possible to know all three components simultaneously; that is, if we

measure the value of one of them (say, the z component), we lose information about the other two components (this effect is related to the uncertainty principle). For this reason, the variance of the z component of angular momentum is zero, meaning we are certain of its value because it has been measured. In other words, there is no distribution of eigenvalues of angular momentum for this component. The opposite situation occurs for the other two components of angular momentum. Once the z component has been measured, the other two components cannot be determined exactly. We know the variance of these two components (see the standard deviation in Eq. (6.69)). The larger the value of m (the projection of angular momentum along the z axis), the smaller the variance and/or standard deviation of the orthogonal components.

Text-based exercises

Solution 6.3. We know that a cross product is given by:

$$\vec{A} \times \vec{B} = (A_x B_y - A_y B_x) \hat{z} + (A_z B_x - A_x B_z) \hat{y} + (A_y B_z - A_z B_y) \hat{x}, \quad (\text{A.138})$$

where the terms in parentheses are the respective vector components of the product under consideration. Our goal is to demonstrate that these components can be written in terms of the Levi-Civita symbol, that is:

$$(\vec{A} \times \vec{B})_i = \epsilon_{ijk} A_j B_k. \quad (\text{A.139})$$

For the particular case $i = x$, for example, we can rewrite the above equation respecting the properties of the Levi-Civita symbol (see Appendix 6.A):

$$(\vec{A} \times \vec{B})_x = \epsilon_{xyz} A_y B_z + \epsilon_{xzy} A_z B_y \quad (\text{A.140})$$

$$= A_y B_z - A_z B_y. \quad (\text{A.141})$$

Note that the result above retrieves that of Eq. (A.138).

Practice exercises

Solution 6.5.

$$\begin{aligned} [\hat{L}_x, \hat{L}_y] &= [(\hat{y}\hat{p}_z - \hat{z}\hat{p}_y), (\hat{z}\hat{p}_x - \hat{x}\hat{p}_z)] \\ &= [\hat{y}\hat{p}_z, \hat{z}\hat{p}_x] - [\hat{y}\hat{p}_z, \hat{x}\hat{p}_z] - [\hat{z}\hat{p}_y, \hat{z}\hat{p}_x] + [\hat{z}\hat{p}_y, \hat{x}\hat{p}_z] \\ &= \hat{y}\hat{p}_x[\hat{p}_z, \hat{z}] + \hat{x}\hat{p}_y[\hat{z}, \hat{p}_z] \\ &= i\hbar(\hat{x}\hat{p}_y - \hat{y}\hat{p}_x) = i\hbar\hat{L}_z. \end{aligned}$$

Solution 6.7.

$$\hat{L}_x = \hbar \begin{pmatrix} 0 & \sqrt{3}/2 & 0 & 0 \\ \sqrt{3}/2 & 0 & 1 & 0 \\ 0 & 1 & 0 & \sqrt{3}/2 \\ 0 & 0 & \sqrt{3}/2 & 0 \end{pmatrix}, \quad (\text{A.142})$$

$$\hat{L}_y = \hbar \begin{pmatrix} 0 & -i\sqrt{3}/2 & 0 & 0 \\ i\sqrt{3}/2 & 0 & -i & 0 \\ 0 & i & 0 & -i\sqrt{3}/2 \\ 0 & 0 & i\sqrt{3}/2 & 0 \end{pmatrix}$$

$$\hat{L}_z = \hbar \begin{pmatrix} 3/2 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & -1/2 & 0 \\ 0 & 0 & 0 & -3/2 \end{pmatrix}, \quad \hat{L}^2 = \frac{15}{4} \hbar^2 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{A.143})$$

Solution 6.8. For an operator to be Hermitian, it must be equal to its conjugate transpose, that is, $\hat{A} = \hat{A}^\dagger = (\hat{A}^T)^*$. It is easy, from the results of the previous problem, to verify that the components of angular momentum have this property (remember that observables are always Hermitian).

On the other hand, the raising and lowering operators are also represented by the matrices:

$$\hat{L}_+ = \hbar \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad \hat{L}_- = \hbar \begin{pmatrix} 0 & 0 & 0 & 0 \\ \sqrt{3} & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}. \quad (\text{A.144})$$

Considering these matrices, it is easy to verify that these raising and lowering operators are non-Hermitian.

Solution 6.9. 1. We can obtain the eigenvalues and the corresponding vectors by solving the eigenvalue equation $\hat{L}_x |\pm\rangle_x = \lambda |\pm\rangle_x$, where λ are the eigenvalues and $|\pm\rangle_x$ the vectors. We also know that the operator $\hat{L}_x$ is represented by the matrix:

$$\hat{L}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{A.145})$$

To avoid trivial solution for the eigenvalue equation, we need to find the determinant of $(\hat{L}_x - \lambda \mathbb{1})$ (where $\mathbb{1}$ is the identity matrix), which reads as: $\lambda_\pm = \pm \hbar/2$. This result is not surprising, because in any direction measuring the angular momentum $l = 1/2$, we must find $\pm \hbar/2$.

2. Each eigenvalue has a corresponding vector; and to determine it, we need to solve the eigenvalue equation again. Therefore, for $\lambda = +\hbar/2$ we

obtain:

$$\frac{\hbar}{2} \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0. \quad (\text{A.146})$$

The solution of this system, along with the normalization condition $|a|^2 + |b|^2 = 1$, leads to the vector:

$$|1/2, +1/2\rangle_x = |+\rangle_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (\text{A.147})$$

In a similar fashion, for the eigenvalue $-\hbar/2$, the corresponding vector is:

$$|1/2, -1/2\rangle_x = |-\rangle_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (\text{A.148})$$

3. We can rewrite the above results in terms of the basis $\{|\pm\rangle\}$ (see Section 6.4 for further details):

$$|+\rangle_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} [|+\rangle + |-\rangle] \quad (\text{A.149})$$

and

$$|-\rangle_x = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} [|+\rangle - |-\rangle]. \quad (\text{A.150})$$

4. From the above results, we can write:

$$\begin{bmatrix} |+\rangle_x \\ |-\rangle_x \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} |+\rangle \\ |-\rangle \end{bmatrix} \quad (\text{A.151})$$

and, consequently:

$$|\pm\rangle_x = \hat{U} |\pm\rangle \Rightarrow |\pm\rangle = \hat{U}^{-1} |\pm\rangle_x \quad (\text{A.152})$$

where $\hat{U}$ is the basis change matrix. Verify that $\hat{U}$ is a unitary matrix.

5. To change the basis of the operator $\hat{L}_x$ using the matrix $\hat{U}$, determined above, we need to refer to the ideas of Subsection 3.A. Thus:

$$\begin{aligned} (\hat{L}_x)_x &= \hat{U} (\hat{L}_x)_z \hat{U}^\dagger \\ &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \end{aligned} \quad (\text{A.153})$$

This operation means a rotation where the x axis becomes the reference axis.

Solution 6.10. The solution to this exercise is absolutely analogous to the previous one; however, the angular momentum operator is different:

$$\hat{L}_y = \frac{\hbar}{2} i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (\text{A.154})$$

1. The eigenvalues are $\lambda_{\pm} = \pm \hbar/2$.
2. The vectors are:

$$|1/2, +1/2\rangle_y = |+\rangle_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad |1/2, -1/2\rangle_y = |-\rangle_y = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \quad (\text{A.155})$$

3. In a similar fashion to the previous case, the relationship between the vectors $|\pm\rangle$ and $|\pm\rangle_y$ can be written as:

$$|+\rangle_y = \frac{1}{\sqrt{2}} [|+\rangle + i|-\rangle] \quad \text{and} \quad |-\rangle_y = \frac{1}{\sqrt{2}} [|+\rangle - i|-\rangle]. \quad (\text{A.156})$$

4. The change of basis matrix can be obtained from the result above:

$$|\pm\rangle_y = \hat{U} |\pm\rangle, \quad (\text{A.157})$$

where

$$\hat{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}. \quad (\text{A.158})$$

Verify that $\hat{U}$ is a unitary matrix.

5. Similarly to the previous case:

$$\begin{aligned} (\hat{L}_y)_y &= \hat{U} (\hat{L}_y)_z \hat{U}^\dagger \\ &= \frac{\hbar}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

Development exercises

Solution 6.11. The procedure for solving this problem is detailed in Exercise 6.9 and, for this reason, let's proceed directly to the final result. To start, we need the operators $\hat{L}_x$ and $\hat{L}_y$, for $l = 1$; which can be found in Eq. (6.83). In this way, we can construct the operator under consideration:

$$\hat{G} = \hbar^2 i \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}. \quad (\text{A.159})$$

1. The eigenvalues of the operator $\hat{G}$ are: $\lambda_1 = 0$, $\lambda_2 = +\hbar^2$ and $\lambda_3 = -\hbar^2$.

2.

$$|\lambda_1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |\lambda_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad |\lambda_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}. \quad (\text{A.160})$$

3. The relationships between the vectors of $\hat{G}$ and the vectors $|\pm\rangle$ can be obtained from the results of the previous item, remembering that the eigenvectors $|\pm\rangle$ were defined in Eq. (6.77). Thus:

$$|\lambda_1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = |1, 0\rangle \quad (\text{A.161})$$

$$|\lambda_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} [-i|1, +1\rangle + |1, -1\rangle] \quad (\text{A.162})$$

$$|\lambda_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} [i|1, +1\rangle + |1, -1\rangle]. \quad (\text{A.163})$$

4. The change of basis matrix can be obtained from the result above:

$$|\lambda_i\rangle = \hat{U}|1, m\rangle, \quad (\text{A.164})$$

where

$$\hat{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & \sqrt{2} & 0 \\ -i & 0 & 1 \\ i & 0 & 1 \end{pmatrix}. \quad (\text{A.165})$$

Verify that $\hat{U}$ is a unitary matrix.

5. Analogous to the case in Exercise 6.9, it is possible to write:

$$\begin{aligned} (\hat{G})_\lambda &= \hat{U}(\hat{G})_z\hat{U}^\dagger \\ &= \hbar^2 \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

Solution 6.12.

$$\begin{aligned} \vec{\hat{L}} \times \vec{\hat{L}} &= (\hat{L}_x\hat{L}_y - \hat{L}_y\hat{L}_x)\hat{z} + (\hat{L}_z\hat{L}_x - \hat{L}_x\hat{L}_z)\hat{y} + (\hat{L}_y\hat{L}_z - \hat{L}_z\hat{L}_y)\hat{x} \\ &= [\hat{L}_x, \hat{L}_y]\hat{z} + [\hat{L}_z, \hat{L}_x]\hat{y} + [\hat{L}_y, \hat{L}_z]\hat{x} \\ &= i\hbar(\hat{L}_z\hat{z} + \hat{L}_y\hat{y} + \hat{L}_x\hat{x}) \\ &= i\hbar\vec{\hat{L}}. \end{aligned} \quad (\text{A.166})$$



It is important to clarify that the symbol $\hat{}$ is used to denote both unit vectors and operators, which, above, appear together in the same equation. Nevertheless, we have maintained this notation to preserve consistency throughout the book. Readers should find it straightforward to distinguish between unit vectors and operators based on the development of the exercise.

Solution 6.13. 1.

$$[\hat{L}^2, \hat{L}^{-2}] = \hat{L}^2 \hat{L}^{-2} - \hat{L}^{-2} \hat{L}^2 = \mathbb{1} - \mathbb{1} = 0 \quad (\text{A.167})$$

where $\mathbb{1}$ is the identity matrix.

2. Since $\hat{L}^2$ and $\hat{L}^{-2}$ commute, we can assert that both are diagonal in the same basis, that is $|l, m\rangle$. In other words, $|l, m\rangle$ are vectors of $\hat{L}^{-2}$. To obtain the eigenvalues of the operator $\hat{L}^{-2}$ we can write:

$$\hat{L}^{-2}|l, m\rangle = \lambda|l, m\rangle. \quad (\text{A.168})$$

Now, we need to perform some operations to determine λ . For this, we start by applying the operator $\hat{L}^2$, by the left, on both sides of the equation above:

$$\hat{L}^2 \hat{L}^{-2}|l, m\rangle = \lambda \left(\hat{L}^2 |l, m\rangle \right). \quad (\text{A.169})$$

We know that:

$$\hat{L}^2 |l, m\rangle = l(l+1)\hbar^2 |l, m\rangle \quad (\text{A.170})$$

and therefore Eq. (A.169) reads as:

$$|l, m\rangle = \lambda l(l+1)\hbar^2 |l, m\rangle. \quad (\text{A.171})$$

Applying a *bra* $\langle l, m|$, by the left, on both sides of the above equation yields:

$$\lambda = \frac{1}{l(l+1)\hbar^2}. \quad (\text{A.172})$$

Thus, the final result reads as:

$$\hat{L}^{-2}|l, m\rangle = [l(l+1)\hbar^2]^{-1}|l, m\rangle. \quad (\text{A.173})$$

Solution 6.14. 1.

$$\begin{aligned} [\hat{r}^2, \hat{L}_u] &= [\hat{x}_v^2, \epsilon_{ulm} \hat{x}_l \hat{p}_m] \\ &= i\hbar \epsilon_{ulm} \{ \hat{x}_v \hat{x}_l \delta_{vm} + \hat{x}_l \delta_{vm} \hat{x}_v \} \\ &= i\hbar \epsilon_{ulm} \{ \hat{x}_m \hat{x}_l + \hat{x}_l \hat{x}_m \} \\ &= 2i\hbar \epsilon_{ulm} \hat{x}_m \hat{x}_l \\ &= 2i\hbar (\hat{x}_m \hat{x}_l - \hat{x}_l \hat{x}_m) = 0. \end{aligned} \quad (\text{A.174})$$

2. This result is expected because purely radial terms (such as $\hat{r}^2$), commute with purely angular terms (such as $\hat{L}_u$).

Solution 6.15. 1.

$$\begin{aligned} [\hat{L}_u, \hat{p}_v] &= [\epsilon_{ujk} \hat{x}_j \hat{p}_k, \hat{p}_v] = \epsilon_{ujk} [\hat{x}_j, \hat{p}_v] \hat{p}_k \\ &= (i\hbar) \epsilon_{ujk} \hat{p}_k \delta_{jv} \\ &= (i\hbar) \epsilon_{uvk} \hat{p}_k. \end{aligned} \quad (\text{A.175})$$

2.

$$\begin{aligned} [\hat{L}_u, \hat{x}_v] &= [\epsilon_{ujk} \hat{x}_j \hat{p}_k, \hat{x}_v] = \epsilon_{ujk} \hat{x}_j [\hat{p}_k, \hat{x}_v] \\ &= (-i\hbar) \epsilon_{ujk} \hat{x}_j \delta_{kv} \\ &= (i\hbar) \epsilon_{uvj} \hat{x}_j. \end{aligned} \quad (\text{A.176})$$

3.

$$[\vec{\hat{A}} \cdot \vec{\hat{L}}, \vec{\hat{B}} \cdot \vec{\hat{L}}] = \delta_{ij} \delta_{uv} [\hat{A}_i \hat{L}_j, \hat{B}_u \hat{L}_v]. \quad (\text{A.177})$$

As the components of the vectors under consideration commute with each other, expanding the above commutator, only one element is non-zero:

$$\begin{aligned} [\vec{\hat{A}} \cdot \vec{\hat{L}}, \vec{\hat{B}} \cdot \vec{\hat{L}}] &= \delta_{ij} \delta_{uv} \hat{A}_i \hat{B}_u [\hat{L}_j, \hat{L}_v] \\ &= \delta_{ij} \delta_{uv} \hat{A}_i \hat{B}_u i\hbar \epsilon_{jvk} \hat{L}_k \\ &= i\hbar (\epsilon_{iuk} \hat{A}_i \hat{B}_u) \hat{L}_k \\ &= i\hbar (\vec{\hat{A}} \times \vec{\hat{B}})_k \hat{L}_k \\ &= i\hbar (\vec{\hat{A}} \times \vec{\hat{B}}) \cdot \vec{\hat{L}} \end{aligned} \quad (\text{A.178})$$

Solution 6.16. The vector diagram of the angular momentum can be seen in Fig. A.3. Note that the figure contains only the information that is known about the angular momentum (magnitude $\sqrt{l(l+1)}\hbar$ and projection along the z axis, i.e., $m\hbar$). Classically, the projection of this vector on the $x-y$ plane is given by:

$$\left(\sqrt{l(l+1)}\hbar \right)^2 = (m\hbar)^2 + r^2 \quad \Rightarrow \quad r = \hbar[l(l+1) - m^2]^{1/2} \quad (\text{A.179})$$

while the standard deviation (of the quantum case) is given by (see Eq. (6.69)):

$$\Delta L_u = \frac{\hbar}{\sqrt{2}} [l(l+1) - m^2]^{1/2} \quad (\text{A.180})$$

where $u = x$ or y . Despite the similarity of the equations, the physics involved is very different. In the classical case, we know all the components

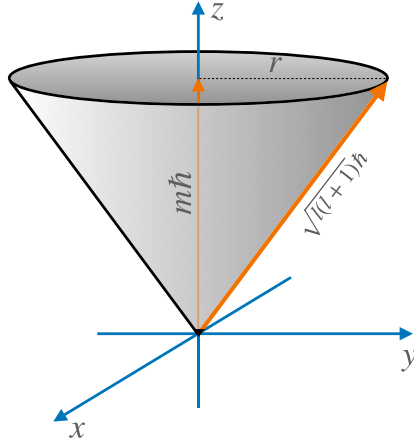


FIGURE A.3 Vector diagram for the angular momentum, with its magnitude and projection along the z -axis. Exercise 6.16.

of the angular momentum, whereas in the quantum case we only know its magnitude and projection along the z -axis, with the standard deviation being one of the few pieces of information we can have about the orthogonal components of the angular momentum.

Solution 6.17. The vector under consideration has a 3×1 dimension and, therefore, represents a system with angular momentum $l = 1$. Thus, there are three possible eigenvalues: ± 1 and 0 (in units of $\hbar$). The probability of obtaining each of these eigenvalues after a measurement of $\hat{L}_x$ is given by:

$$P(l, m) = |{}_x\langle l, m | \Psi \rangle|^2. \quad (\text{A.181})$$

From Example 6.2, we know the vectors $|1, \pm 1\rangle_x$ and $|1, 0\rangle_x$ and thus it is possible to calculate:

$$P(1, +1) = \left| \frac{1}{2} \begin{pmatrix} 1 & \sqrt{2} & 1 \end{pmatrix} \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right|^2 = \frac{|2 + \sqrt{2}|^2}{14} \quad (\text{A.182})$$

$$P(1, -1) = \left| \frac{1}{2} \begin{pmatrix} 1 & -\sqrt{2} & 1 \end{pmatrix} \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right|^2 = \frac{|2 - \sqrt{2}|^2}{14} \quad (\text{A.183})$$

$$P(1, 0) = \left| \frac{1}{2} \begin{pmatrix} \sqrt{2} & 0 & -\sqrt{2} \end{pmatrix} \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right|^2 = \frac{1}{7}. \quad (\text{A.184})$$

Solution 6.18. The solution procedure is in Example 6.4. The final answer is:

$$Y_2^{+2}(\theta, \phi) = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \sin^2(\theta) e^{2i\phi}. \quad (\text{A.185})$$

Solution 6.19. Considering:

$$\hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-) \quad \text{and} \quad \hat{L}_y = \frac{1}{2i}(\hat{L}_+ - \hat{L}_-), \quad (\text{A.186})$$

it is possible to write:

$$\begin{aligned} \hat{L}_x \hat{L}_y &= \frac{1}{4i}(\hat{L}_+ + \hat{L}_-)(\hat{L}_+ - \hat{L}_-) \\ &= \frac{1}{4i}(\hat{L}_+^2 - \hat{L}_-^2 - 2\hbar \hat{L}_z). \end{aligned} \quad (\text{A.187})$$

Our goal is to determine the expectation value $\langle \hat{L}_x \hat{L}_y \rangle$. It is easy to see that $\langle \hat{L}_\pm^n \rangle = 0$ (where n is an integer), since this expectation value is proportional to $\langle l, m | l, m \pm n \rangle = 0$. Thus:

$$\langle \hat{L}_x \hat{L}_y \rangle = i \frac{m\hbar^2}{2}. \quad (\text{A.188})$$

Similarly to the procedure above, we can conclude that:

$$\langle \hat{L}_y \hat{L}_x \rangle = -i \frac{m\hbar^2}{2}. \quad (\text{A.189})$$

Solution 6.20. Analogous to the previous exercise, we can write:

$$\hat{L}_x \hat{L}_z = \frac{1}{2}(\hat{L}_+ \hat{L}_z + \hat{L}_- \hat{L}_z), \quad (\text{A.190})$$

which leads to $\langle \hat{L}_x \hat{L}_z \rangle = 0$, due to the fact that the raising and lowering operators lead to $\langle l, m | l, m \pm 1 \rangle = 0$. Based on similar arguments: $\langle \hat{L}_y \hat{L}_z \rangle = 0$.

Advanced exercises

Solution 6.21. 1.

$$\hat{S}_i = \frac{\hbar}{2} \hat{\sigma}_i \quad (\text{A.191})$$

where $\hat{\sigma}_i$ are the Pauli matrices ($i = x, y, z$):

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (\text{A.192})$$

2. The operator $\hat{S}_n$ is the one in the direction $\hat{n}$ and, therefore:

$$\hat{S}_n = \hat{n} \cdot \vec{\hat{S}} = \frac{\hbar}{2} \begin{pmatrix} \cos \theta & e^{-i\phi} \sin \theta \\ e^{i\phi} \sin \theta & -\cos \theta \end{pmatrix}. \quad (\text{A.193})$$

3. We know that the eigenvalues of any spin $s = 1/2$ operator are always $\pm \hbar/2$, but let's calculate! The determinant of the eigenvalue equation for $\hat{S}_n$ must be zero, and, therefore:

$$\begin{vmatrix} \hbar \cos(\theta)/2 - \lambda & \hbar e^{-i\phi} \sin(\theta)/2 \\ \hbar e^{i\phi} \sin(\theta)/2 & -(\hbar \cos(\theta)/2 + \lambda) \end{vmatrix} = 0 \Rightarrow \lambda_{\pm} = \pm \frac{\hbar}{2}. \quad (\text{A.194})$$

4. For $\lambda = +\hbar/2$ we need to solve the eigenvalue equation to obtain the vector elements. Thus:

$$\begin{pmatrix} \cos \theta - 1 & e^{-i\phi} \sin \theta \\ e^{i\phi} \sin \theta & -(\cos \theta + 1) \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = 0, \quad (\text{A.195})$$

which, after solving and considering the normalization condition $|a|^2 + |b|^2 = 1$ yields:

$$|+\rangle_n = \begin{pmatrix} e^{-i\phi} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix}. \quad (\text{A.196})$$

Analogously, for $\lambda = -\hbar/2$, the vector reads as:

$$|-\rangle_n = \begin{pmatrix} -e^{-i\phi} \sin(\theta/2) \\ \cos(\theta/2) \end{pmatrix}. \quad (\text{A.197})$$

5. Initially, the system is in the state $|\Psi\rangle = |+\rangle$ and therefore, the probability of obtaining each eigenvalue can be written as:

$$P(+) = |{}_n\langle +|\Psi\rangle|^2 = \cos^2(\theta/2) \quad \text{and} \quad P(-) = |{}_n\langle -|\Psi\rangle|^2 = \sin^2(\theta/2). \quad (\text{A.198})$$

Note that these probabilities have azimuthal symmetry, depending only on the polar angle θ . Think about this result!

Solution 6.22. The solution procedure is in Example 6.5. The final answer is:

$$P(1, \pm 1) = 1/2 \quad \text{and} \quad P(1, 0) = 0. \quad (\text{A.199})$$

Solution 6.23. Let's start with the definition of angular momentum $\vec{\hat{L}} = \vec{r} \times \vec{\hat{p}}$. Considering the Hamiltonian has a term proportional to $\hat{p}^2$, it is possible to substitute it by a term containing $\hat{L}^2 = (\vec{r} \times \vec{\hat{p}}) \cdot (\vec{r} \times \vec{\hat{p}})$. The components of this last vector can be written as:

$$\begin{aligned}
\hat{L}_i^2 &= \epsilon_{ijk} \epsilon_{ilm} \hat{x}_j \hat{p}_k \hat{x}_l \hat{p}_m \\
&= (\delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}) \hat{x}_j \hat{p}_k \hat{x}_l \hat{p}_m \\
&= \hat{x}_j^2 \hat{p}_k^2 - \hat{x}_j \hat{p}_j \hat{x}_k \hat{p}_k,
\end{aligned} \tag{A.200}$$

which becomes $\hat{L}^2 = r^2 \hat{p}^2 - (\vec{r} \cdot \vec{\hat{p}})^2$. Therefore, we obtain a relationship between $\hat{p}^2$ and $\hat{L}^2$, which can be rewritten as:

$$\hat{p}^2 = \frac{\hat{L}^2 + (\vec{r} \cdot \vec{\hat{p}})^2}{r^2}. \tag{A.201}$$

From the above result, we can rewrite the Hamiltonian of a three-dimensional harmonic oscillator in terms of the angular momentum operator:

$$\hat{\mathcal{H}} = \frac{\hat{p}^2}{2m} + \frac{m}{2} \omega^2 r^2 \Rightarrow \hat{\mathcal{H}} = \frac{\hat{L}^2}{2mr^2} + \frac{(\vec{r} \cdot \vec{\hat{p}})^2}{2mr^2} + \frac{m}{2} \omega^2 r^2. \tag{A.202}$$

However, the problem requests to write the Hamiltonian in terms of the angular momentum operator (task accomplished above), and radial contributions only. Therefore, we need to consider the differential form of the linear momentum operator $\vec{\hat{p}} = -i\hbar \vec{\nabla}$. Considering:

$$\vec{\nabla} = \hat{r} \frac{\partial}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \tag{A.203}$$

we obtain:

$$\vec{r} \cdot \vec{\hat{p}} = -i\hbar r \frac{\partial}{\partial r} \Rightarrow (\vec{r} \cdot \vec{\hat{p}})^2 = -\hbar^2 r \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right). \tag{A.204}$$

From the above, we obtain the final Hamiltonian:

$$\hat{\mathcal{H}} = \frac{\hat{L}^2}{2mr^2} - \frac{\hbar^2}{2m} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{m}{2} \omega^2 r^2. \tag{A.205}$$

2. Eq. (A.205) above has three terms: one depends only on $\hat{L}^2$, but we know that $[\hat{L}^2, \hat{L}_z] = 0$. The other two terms have only radial contributions, which therefore also commute with $\hat{L}_z$. Thus, we can state that $[\hat{\mathcal{H}}, \hat{L}_z] = 0$.

A.7 Chapter 7

Conceptual exercises

Solution 7.1. One reason to expand the Hilbert space is to enable mathematical operations involving matrices that represent operators of different dimensions. For example, consider the Hamiltonian $\hat{\mathcal{H}} = \hat{S}_z^2$ (where